

With a Boltzmann distribution of incident particles

TABLE I
COMPUTED BARRIER PENETRATION QUANTUM CORRECTIONS Γ^* FROM UNSYMMETRICAL ECKART BARRIERS AS A FUNCTION
OF α_1 , α_2 , AND u^*

α_1	α_2	u^*								
		2	3	4	5	6	8	10	12	16
0.5	0.5	1.16	1.25	1.34	1.44	1.55	1.80	2.09	2.42	3.26
	1	1.13	1.21	1.29	1.38	1.47	1.68	1.93	2.22	2.94
	2	1.09	1.14	1.20	1.27	1.34	1.51	1.71	1.94	2.53
	4	1.04	1.07	1.11	1.16	1.22	1.35	1.50	1.69	2.16
	8	0.99	1.00	1.03	1.06	1.11	1.21	1.34	1.49	1.88
	12	0.96	0.97	0.99	1.02	1.06	1.15	1.26	1.40	1.76
	16	0.94	0.95	0.97	0.99	1.02	1.11	1.22	1.35	1.68
1	20	0.93	0.94	0.95	0.97	1.00	1.08	1.19	1.31	1.64
	1	1.27	1.43	1.62	1.83	2.09	2.72	3.56	4.68	8.19
	2	1.21	1.35	1.51	1.71	1.93	2.50	3.26	4.28	7.48
	4	1.14	1.24	1.37	1.53	1.71	2.16	2.78	3.60	6.16
	8	1.08	1.16	1.26	1.39	1.54	1.92	2.43	3.12	5.25
	12	1.06	1.12	1.21	1.33	1.46	1.81	2.28	2.91	4.88
	16	1.04	1.10	1.18	1.29	1.42	1.75	2.20	2.80	4.66
2	20	1.03	1.08	1.16	1.26	1.39	1.70	2.14	2.72	4.52
	2	1.32	1.58	1.91	2.34	2.90	4.55	7.34	12.1	34.0
	4	1.26	1.47	1.77	2.16	2.66	4.20	6.85	11.4	33.4
	8	1.19	1.36	1.61	1.93	2.36	3.65	5.87	9.69	28.0
	12	1.16	1.32	1.54	1.84	2.23	3.41	5.44	8.94	25.6
	16	1.14	1.29	1.50	1.78	2.15	3.27	5.20	8.51	24.2
	20	1.12	1.27	1.47	1.74	2.10	3.18	5.03	8.22	23.3
4	4	1.30	1.58	2.02	2.69	3.69	7.60	17.3	42.4	304
	8	1.25	1.51	1.93	2.56	3.56	7.57	18.0	46.7	376
	12	1.22	1.47	1.86	2.46	3.39	7.16	17.0	44.0	354
	16	1.20	1.44	1.81	2.39	3.28	6.88	16.2	41.9	335
	20	1.19	1.42	1.78	2.34	3.20	6.68	15.7	40.3	321
	8	1.24	1.56	2.04	2.94	4.54	13.8	57.0	307	..
	12	1.22	1.54	2.04	2.96	4.68	15.4	71.7	445	..
8	16	1.21	1.53	2.02	2.93	4.65	15.6	74.4	473	..
	20	1.20	1.51	2.00	2.90	4.61	15.5	74.2	474	..
	12	1.2	1.5	2.1	3.1	5.2	22	162	1970	..
	16	1.2	1.5	2.1	3.1	5.4	25	220	3300	..
	20	1.2	1.5	2.1	3.1	5.4	26	246	3920	..
	16	1.2	1.5	2.1	3.2	5.7	32	437	...	..
	20	1.2	1.5	2.1	3.2	5.9	37	616	...	..
20	20	1.2	1.5	2.1	3.2	6.1	46	1150	...	..

at $-\infty$ in Fig. 1, the correction factor can be expressed as

$$\Gamma^* = e^{\Delta V_1/kT} \int_0^\infty e^{-E/kT} \kappa(E) d(E/kT) \quad (18)$$

With the transmission function given by eq. 13, we have numerically integrated eq. 18 by means of the IBM 704 computer at the Berkeley Computing Center. The results can be expressed in terms of α_1 , α_2 and u^* , eq. 9-12. For chemically interesting values of these parameters, the quantum correction factors are listed in Table I. It can be seen that for a given α_1 and u^* , the tunnelling correction is moderately sensitive to the value of α_2 . One would make a fairly large error, in most cases, if one used the symmetrical Eckart correction^{3,4} for an unsymmetrical case. It should be further noted⁵

that if α_1 and α_2 are interchanged, the value of Γ^* is unaffected.

In ref. 4 and 6, it was emphasized that for chemical reactions, ΔV_1 is *not* the activation energy. The one-dimensional reaction coordinate does not extend from the activated complex to the products or reactants. The reaction normal mode, when extended beyond the quadratic region near the saddlepoint, soon encounters "side-wall-repulsions" on the two-dimensional potential energy surface. In fitting an Eckart potential to a potential energy surface for a chemical reaction, $-\infty$ is not at reactants and $+\infty$ is not at products; rather in terms of the potential energy surface, $-\infty$ is a cirque immediately adjacent to the col and $+\infty$ is a similar cirque on the other side of the col.

(6) H. S. Johnston, *Advances in Chem. Phys.*, **3**, 131 (1961).